By hand calculations

22 november 2024

States

The state space is defined as:

$$O = \{(C,C)\}, \quad 1 = \{(D,C),(C,D),(D,D)\}$$

Payoff Matrix

The payoff matrix for the game is:

	C	D
C	$R = 0.8$	$S = 0$
D	$T = 1$	$P = 0.2$

Q-value Updates for Opponent E-GT

For 0:

1. Cooperation in 0:

$$Q_{00} = (1 - \frac{\varepsilon}{2})\Big[R + \delta \max\{Q_{00}, Q_{10}\}\Big] + \frac{\varepsilon}{2}\Big[S + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

2. Defection in 0:

$$Q_{01} = (1 - \frac{\varepsilon}{2})\Big[T + \delta \max\{Q_{01}, Q_{11}\}\Big] + \frac{\varepsilon}{2}\Big[P + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

For 1:

1. Cooperation in 1:

$$Q_{10} = (1 - \frac{\varepsilon}{2})\Big[S + \delta \max\{Q_{00}, Q_{10}\}\Big] + \frac{\varepsilon}{2}\Big[R + \delta \max\{Q_{00}, Q_{01}\}\Big]$$

2. Defection in 1:

$$Q_{11} = (1 - \frac{\varepsilon}{2})\Big[P + \delta \max\{Q_{01}, Q_{11}\}\Big] + \frac{\varepsilon}{2}\Big[T + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

1 Scenarios for Different Opponent Strategies

1.1 Case 1: Playing GT

If the opponent plays $\varepsilon - GT$ and I play $\varepsilon - GT$, the Q-values are updated as:

$$Q_{00} = (1-\frac{\varepsilon}{2})\Big[R+\delta Q_{00}\Big] + \frac{\varepsilon}{2}\Big[S+\delta Q_{11}\Big]$$

$$Q_{01} = (1-\frac{\varepsilon}{2})\Big[T+\delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[P+\delta Q_{11}\Big]$$

$$Q_{10} = (1-\frac{\varepsilon}{2})\Big[S+\delta Q_{10}\Big] + \frac{\varepsilon}{2}\Big[R+\delta Q_{00}\Big]$$

$$Q_{11} = (1-\frac{\varepsilon}{2})\Big[P+\delta Q_{01}\Big] + \frac{\varepsilon}{2}\Big[T+\delta Q_{11}\Big]$$

1.1.1 Solving this system of equations

The equations for Q_{ij} are derived as follows:

Equation for Q_{00}:

$$\begin{aligned}
Q_{00} &= (1-\frac{\varepsilon}{2})(R+\delta Q_{00}) + \frac{\varepsilon}{2}(S+\delta Q_{11}) \\
Q_{00} &= (1-\frac{\varepsilon}{2})R + (1-\frac{\varepsilon}{2})\delta Q_{00} + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{00} - (1-\frac{\varepsilon}{2})\delta Q_{00} &= (1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{00}\Big[1-(1-\frac{\varepsilon}{2})\delta\Big] &= (1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{00} &= \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}}{1-(1-\frac{\varepsilon}{2})\delta}
\end{aligned}$$

Equation for Q_{01}:

$$\begin{aligned}
Q_{01} &= (1-\frac{\varepsilon}{2})(T+\delta Q_{11}) + \frac{\varepsilon}{2}(P+\delta Q_{11}) \\
Q_{01} &= (1-\frac{\varepsilon}{2})T + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}P + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{01} &= (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta Q_{11}
\end{aligned}$$

Equation for Q_{10}:

$$\begin{aligned}
Q_{10} &= (1-\frac{\varepsilon}{2})(S+\delta Q_{10}) + \frac{\varepsilon}{2}(R+\delta Q_{00}) \\
Q_{10} &= (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta Q_{10} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00} \\
Q_{10} - (1-\frac{\varepsilon}{2})\delta Q_{10} &= (1-\frac{\varepsilon}{2})S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00} \\
Q_{10}\Big[1-(1-\frac{\varepsilon}{2})\delta\Big] &= (1-\frac{\varepsilon}{2})S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00} \\
Q_{10} &= \frac{(1-\frac{\varepsilon}{2})S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00}}{1-(1-\frac{\varepsilon}{2})\delta}
\end{aligned}$$

Equation for Q_{11}:

$$\begin{aligned}
Q_{11} &= (1-\frac{\varepsilon}{2})\,(P+\delta Q_{11}) + \frac{\varepsilon}{2}\,(T+\delta Q_{11}) \\
Q_{11} &= (1-\frac{\varepsilon}{2})P + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}T + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{11} - [(1-\frac{\varepsilon}{2})\delta + \frac{\varepsilon}{2}\delta]Q_{11} &= (1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T \\
Q_{11}\,[1-\delta] &= (1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T \\
Q_{11} &= \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}
\end{aligned}$$

Final System of Equations:

$$\begin{aligned}
Q_{00} &= \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}}{1-(1-\frac{\varepsilon}{2})\delta} \\
Q_{01} &= (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta Q_{11} \\
Q_{10} &= \frac{(1-\frac{\varepsilon}{2})S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00}}{1-(1-\frac{\varepsilon}{2})\delta} \\
Q_{11} &= \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}
\end{aligned}$$

Substituting one in another: The reduced system of equations is given as follows:

$$Q_{00} = \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\frac{\delta\left[(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T\right]}{1-\delta}}{1-(1-\frac{\varepsilon}{2})\delta},$$

$$Q_{01} = (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \frac{\delta\left[(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T\right]}{1-\delta},$$

$$Q_{10} = \frac{(1-\frac{\varepsilon}{2})S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta\cdot\frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\frac{\delta\left[(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T\right]}{1-\delta}}{1-(1-\frac{\varepsilon}{2})\delta}}{1-(1-\frac{\varepsilon}{2})\delta},$$

$$Q_{11} = \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}.$$